

Problem Description - Smart Digits

Smart digits are the new types of digits created by Chapeco researchers. It is the value of the sum of digits of all numbers within a given range. For a small range, for example from 1 to 5, the sum of the digits (1,2,3,4,5) is 15. This they can solve easily. But for a larger range they cannot calculate the sum easily and have asked for your help. Can you help them? Specifically, if given two integers L and R, find the sum of all the digits of every number in this range. Since the number can be very large return the answer as modulus $10^9 + 7$.

For example, given L = 10 and R = 15, the sum of the digits of the numbers 10,11,12,13,14,15 will be 21 $((1+0) + (1+1) + (1+2) + (1+3) + (1+4) + (1+5))$.

Input

The first line of input gives the number of test cases (T). The next T gives specify two numbers L and R, where $1 \leq L, R \leq 10^{18}$.

Output

For each L and R pair, print the sum of the digits in the interval L, R as described above.

Input Sample	Output Sample
4 1 5 10 15	15 21

Methods that have to be used in your solution

1	<code>int sumOfRange(long a, long b)</code> - This method should be called by the main() method. It should calculate the sum of all numbers in this range from a to b and return the sum to the main method()
2	<code>int sumOfDigits(int a)</code> - This method should be called by the sumOfRange() method. It should calculate and return the sum of all digits of number a

Marking Rubric				
Name of Method	Poor	Average	Excellent	Marks obtained
<code>main()</code>	0	1.5	3	
<code>sumOfRange(int a, int b)</code>	0	1.5	3	
<code>sumOfDigits(int a)</code>	0	2.0	4	
Total out of 10				

دفعه ۲۱.۵ / نموذج ۲

Problem Description – Reversed Digits

Reversed digits is the value of the sum of the reversed digits of all numbers within a given range. For a small range, for example from 3 to 8, the sum of the digits (8,7,6,5,4,3) is 33. This they can solve easily. But for a larger range they cannot calculate the sum easily and have asked for your help. Can you help them? Specifically, if given two integers A and B, find the sum of all the digits in reverse of every number in this range. Since the number can be very large return the answer as modulus $10^9 + 7$.

For example, given $L = 24$ and $R = 29$, the sum of the digits of the numbers 29, 28, 27, 26, 25, 24 will be 51 $\{(2+9) + (2+8) + (2+7) + (2+6) + (2+5) + (2+4)\}$.

Input

The first line of input gives the number of test cases (T). The next T lines specify two numbers L and R, where $1 \leq L, R \leq 10^{18}$.

Output

For each L and R pair, print the sum of the digits in the interval L, R as described above.

Input Sample	Output Sample
2 1 8 24 29	33 51

Methods that have to be used in your solution

1	<code>int sumOfReversedDigits(int x)</code> - This method should be called by the <code>reverseRange()</code> method. It should calculate and return the sum of all digits of number x.
2	<code>int reverseRange(long x, long y)</code> - This method should be called by the <code>main()</code> method. It should calculate the sum of all numbers in this range from x to y and return the sum to the main method()

Marking Rubric				
Name of Method	Poor	Average	Excellent	Marks obtained
<code>main()</code>	0	1.5	3	
<code>reverseRange(int a, int b)</code>	0	1.5	3	
<code>sumOfReversedDigits(int x)</code>	0	2.5	4	
Total out of 10				

